The compact G_2 sigma-MGF on V_7

Proof and exact low-degree matrix verification

Computational proof companion

17 July 2026

Abstract

We prove the compact Molien–Weyl formula for the minimal seven-dimensional G_2-module, construct its maximal-torus matrices from the weights, and verify every Schur coefficient through degree three exactly. The result is $\mathcal{S}_{G_2,7} = 1 + h_2 + e_3 + O_{\geq 4}$, and the ordinary specialization is $(1-t^2)^{-1}$.

1 General compact-group identity

For a unitary K-module V and eigenvalues $\lambda_j(g)$, the Cauchy identity gives, as a formal power series,

$$\prod_{a,j} (1 - t_a \lambda_j(g))^{-1} = \sum_{\lambda} s_\lambda(\mathbf{t})\, \chi_{\mathbb{S}_\lambda V}(g).$$

Haar integration projects each representation onto its invariant subspace; hence

$$\boxed{\mathcal{S}_{K,V}(\mathbf{t}) = \int_K \prod_a \det(I - t_a \rho(g))^{-1}\, dg = \sum_\lambda \dim(\mathbb{S}_\lambda V)^K s_\lambda(\mathbf{t}).}$$

Expanding each determinant separately also gives the requested monomial form

$$[m_\tau]\mathcal{S}_{K,V} = \dim \left(\bigotimes_i \operatorname{Sym}^{\tau_i} V \right)^K.$$

If T is a maximal torus, Weyl integration yields

$$\mathcal{S}_{K,V} = \frac{1}{|W|} \operatorname{CT}_z \left[\prod_{\alpha > 0} (1 - z^\alpha)(1 - z^{-\alpha}) \prod_{a,\mu} (1 - t_a z^\mu)^{-m(\mu)} \right].$$

This proves the proposed compact analogue of the finite Molien formula.

2 Specialization to the minimal module

The weights of V_7 are zero together with the six short roots. Since $|W(G_2)| = 12$,

$$\boxed{\mathcal{S}_{G_2,7} = \frac{1}{12} \operatorname{CT}_z \left[D_{G_2}(z) \prod_a \left((1 - t_a)^{-1} \prod_{\beta \in \Phi_{\text{short}}} (1 - t_a z^\beta)^{-1} \right) \right]}.$$

The invariant metric lies in $\operatorname{Sym}^2 V_7$, while the octonionic associative form lies in $\bigwedge^3 V_7$. Their uniqueness and the absence of other invariants below degree four give

$$\boxed{\mathcal{S}_{G_2,7} = 1 + h_2 + e_3 + O_{\geq 4}.}$$

A standard invariant-theory theorem for the minimal G_2 module states that $\mathbb{C}[V_7]^{G_2} = \mathbb{C}[q]$, where q is the invariant quadratic norm. Using that external theorem, $\mathcal{S}_{G_2,7}(t, 0, \ldots) = (1-t^2)^{-1}$. The exact Weyl computation in this note independently verifies the coefficients through degree three.

3 Exact matrix verification

The notebook generates the root system from the G_2 Cartan matrix and forms

$$M(\theta) = \operatorname{diag}\big(e^{i\langle\mu,\theta\rangle}\big)_{\mu\in\{0\}\cup\Phi_{\text{short}}} \in U(7).$$

It compares $\prod_a \det(I - t_a M)^{-1}$ directly with the product over the seven diagonal eigenvalues. For integration it evaluates Schur characters by Jacobi–Trudi and uses the one-sided Weyl denominator

$$\prod_{\alpha>0}(1 - z^{-\alpha}) = \sum_{w\in W} \det(w) z^{w\rho-\rho}.$$

Thus every coefficient is an exact integer, with no quadrature tolerance. In the requested monomial basis the exact degree-at-most-three signature is

$$([m_{(2)}], [m_{(1,1)}], [m_{(3)}], [m_{(2,1)}], [m_{(1,1,1)}]) = (1, 1, 0, 0, 1).$$

λ	$\varnothing$	(1)	(2)	$(1,1)$	(3)	$(2,1)$	$(1,1,1)$
proposed	1	0	1	0	0	0	1
exact Weyl check	1	0	1	0	0	0	1

The sampled torus matrix has unitarity residual 3.2×10^{-16} and the two integrand evaluations differ by less than 9×10^{-16}. All seven coefficient checks pass.

Reproducibility. Run `g2_verification.py` with marimo. The shared exact-arithmetic module is in the parent folder.

External invariant-theory source

See J. F. Adams, *Lectures on Exceptional Lie Groups*, Chicago Lectures in Mathematics (1996), for the minimal G_2 module and its invariant metric. The all-degree one-copy invariant-ring statement is an external input; the low-degree Weyl checks above are internal to this artifact.